

Task3

OPTION A

1. Project Summary

In this project, since real Goldman Sachs credit approval data involves trade secrets, customer privacy, and regulatory restrictions and cannot be publicly obtained, I used publicly available German credit data as alternative data to conduct a prototype simulation study on the compliance and fairness issues that Goldman Sachs may face in the credit scoring scenario.

[Statlog \(German Credit Data\)](#)

This project is a tool for cross-jurisdictional credit scoring compliance audits. Its goal is to use the same model to output different fairness audit conclusions under the two regulatory logics of US Regulation B and EU AI Act. In my code, the US rule checks Disparate Impact Ratio ($DIR \geq 0.80$), while the EU rule checks Demographic Parity Difference ($DPD \leq 0.05$).

The program uses the German Credit Dataset as input, generates credit scores based on Logistic Regression, and translates regulatory rules into a jurisdiction configuration layer: the US uses the disparate impact ratio as the primary indicator, while the EU uses the demographic parity difference.

In my baseline results, the model's DIR was 0.9155 and DPD was 0.0608, meaning the same model passed in the US but failed in the EU, demonstrating that the tool truly captures cross-jurisdictional rule discrepancies.

I added a threshold sensitivity analysis to test the robustness of the model, with the threshold representing the stringency of loan approval. Within the quantile range of 0.20 to 0.80, the US remained “qualified” across all listed settings, while the EU’s status fluctuated between “qualified” and “unqualified” depending on the threshold. This indicates that the fairness conclusion is highly sensitive to policy choices and operational thresholds, thus requiring regular monitoring.

In addition to basic auditing, I also included a dashboard, threshold sensitivity analysis, SHAP explanation, and model card stub to demonstrate how this prototype supports compliance interpretation, management communication. This tool should support human governance decisions, not replace legal or compliance judgment.

2. Senior Management Pitch (Contains Short Deck)

Why is this product right for our company?

As a globally operating financial institution, we face the risk of differing regulatory rules across countries when expanding our lending business. This product perfectly aligns with our globalization strategy: at the "decision level," it achieves localized compliance through a flexible jurisdictional rules engine, thus striking an optimal balance between "technology scaling" and "compliance localization." It also analyzes credit scores, which can be used to explain the reasons to regulators and consumers.

What problem does this tool solve that a memo or spreadsheet cannot?

Memos or Excel spreadsheets can only provide static compliance documentation, failing to measure dynamic, genuine risk. When business units want to fine-tune credit approval thresholds to boost revenue, spreadsheets cannot simulate the differential impact of such adjustments on fairness metrics in real time. Our tool's threshold sensitivity analysis can dynamically calculate compliance boundaries at different quantiles. Furthermore, spreadsheets cannot provide individual interpretations, while our tool can specifically analyze the exact reasons for rejection.

What regulatory divergence does it handle, and how?

The jurisdiction rules are parameterized, for example, the US uses `disparate_impact_ratio >= 0.80` and the EU uses `demographic_parity_difference <= 0.05`. Then, the same model output is used to enter different rule engines, resulting in different audit statuses. If more regions need to be adapted in the future, their specific rules only need to be input into `JURISDICTION_CONFIG`.

What does it not do — and why did you make that boundary choice?

This tool is only responsible for "Audit & Alert," and will not alter, auto-reweigh, or automatically modify model parameters to force compliance.

This is because "judgment cannot be outsourced." Fairness testing is a trade-off between business and ethics; blindly automating fixes could introduce reverse discrimination or significantly sacrifice the model's predictive accuracy, thereby increasing the risk of credit default. We choose to keep the tool at the judgment stage, returning the final corrective power and responsibility to the official body.